

Fundamentals of Information Systems Security

Chapter 7 Quiz: Malicious Software and Attack Vectors

Name: _____

Date: _____

Total Points: 100

Time Allowed: 30 Minutes

Instructions

- This quiz contains **20 questions** covering Chapter 7
- Answer all questions to the best of your ability
- Multiple choice questions have only one correct answer
- True/False questions should be marked T or F
- Short answer questions require concise, accurate responses

Section A: Multiple Choice (40 points)

1. (4 points) What is malware?
 - A. Software that performs actions contrary to the user's intentions
 - B. Software that improves system performance
 - C. Software that is always harmless
 - D. Software that only affects Windows systems
2. (4 points) Which type of malware requires user action to spread?
 - A. Worm
 - B. Virus
 - C. Trojan
 - D. Ransomware
3. (4 points) What is a rootkit?
 - A. A type of virus that infects boot sectors
 - B. Malware that modifies or replaces programs to hide compromise
 - C. A network-based attack tool
 - D. A type of encryption software
4. (4 points) Which type of malware encrypts files and demands payment for decryption?
 - A. Virus
 - B. Worm
 - C. Ransomware

- D. Spyware
5. (4 points) What is the difference between a virus and a worm?
- A. Viruses require user action; worms self-propagate
 - B. Worms require user action; viruses self-propagate
 - C. They are the same thing
 - D. Viruses only affect Windows; worms affect Linux
6. (4 points) Which of the following is a social engineering attack?
- A. SQL Injection
 - B. Buffer Overflow
 - C. Phishing
 - D. Cross-Site Scripting
7. (4 points) What is a botnet?
- A. A network of compromised computers controlled by an attacker
 - B. A type of antivirus software
 - C. A network security device
 - D. A type of encryption algorithm
8. (4 points) What type of attack floods a target with traffic to make it unavailable?
- A. Man-in-the-Middle
 - B. Denial-of-Service (DoS)
 - C. SQL Injection
 - D. Cross-Site Scripting
9. (4 points) What is the first phase of a cyber attack?
- A. Weaponization
 - B. Delivery
 - C. Reconnaissance
 - D. Exploitation
10. (4 points) What is a honeypot?
- A. A decoy system designed to attract attackers
 - B. A type of firewall
 - C. A vulnerability scanner
 - D. A password cracker

Section B: True/False (20 points)

11. (2 points) Trojan horses do not self-replicate.
- A. True
 - B. False
12. (2 points) Polymorphic viruses change their appearance with each infection.

- A. True
 - B. False
13. (2 points) Spam is estimated to account for 10-20
- A. True
 - B. False
14. (2 points) A logic bomb executes when specific conditions are met.
- A. True
 - B. False
15. (2 points) Keyloggers only capture keystrokes and cannot capture screenshots.
- A. True
 - B. False
16. (2 points) SQL injection is a type of injection attack.
- A. True
 - B. False
17. (2 points) Defense-in-depth means using only one security control.
- A. True
 - B. False
18. (2 points) Zero-day vulnerabilities have no patch available when discovered.
- A. True
 - B. False
19. (2 points) Spear phishing is a generalized, untargeted phishing attack.
- A. True
 - B. False
20. (2 points) War driving is a technique for locating wireless networks.
- A. True
 - B. False

Section C: Short Answer (40 points)

21. (10 points) Explain the difference between a virus, worm, and Trojan horse. How does each spread?

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22. (10 points) Describe the five phases of a structured cyber attack from reconnaissance to covering tracks.
23. (10 points) What is defense-in-depth? List at least four layers of defense.
24. (10 points) Explain the difference between phishing, spear phishing, and pharming. Which is most dangerous and why?

End of Quiz